

Part A: Literature Review and Public Dataset

CAP182 Capstone Project: STADIOEquities

Publication 1

Title	<i>A data-driven approach to predict the success of bank telemarketing</i>
Harvard reference	Moro, S., Cortez, P. and Rita, P. (2014) 'A data-driven approach to predict the success of bank telemarketing', <i>Decision Support Systems</i> , 62, pp. 22–31. doi: 10.1016/j.dss.2014.03.001.
Link	https://doi.org/10.1016/j.dss.2014.03.001 Open-access post-print: https://repositorio.iscte-iul.pt/bitstream/10071/9499/5/dss_v3.pdf
Modelling	The study compared logistic regression, decision trees, neural networks and support vector machines (SVM). Model performance was evaluated using the area under the ROC curve (AUC) and cumulative LIFT curves. A rolling time-window evaluation approach was also used.
Preprocessing techniques	The study initially considered a large set of 150 candidate features covering client demographic information, campaign-contact information and macroeconomic variables. A semi-automatic feature selection step reduced these to a set of 22 relevant features. The study also used a rolling time-window approach to evaluate model performance across different periods.
Datasets used	Portuguese bank telemarketing campaign data. The dataset was subsequently made publicly available through the UCI Machine Learning Repository as the Bank Marketing dataset.
Key takeaways	1. Customer and campaign-related information can be useful for predicting financial-product conversion outcomes. 2. Time-aware evaluation can be important when working with data collected over different periods. This project will consider a time-ordered split when evaluating its models. 3. Comparing several modelling techniques can help identify an appropriate approach rather than relying on a single model. This project will therefore compare two different models. Relevance to STADIOEquities: The study is relevant because its binary prediction of whether a customer converts to a financial product provides a useful comparison for predicting whether a newly registered customer will make a first deposit.

Publication 2

Title	<i>Investigating customer churn in banking: a machine learning approach and visualization app for data science and management</i>
Harvard reference	Singh, P.P., Anik, F.I., Senapati, R., Sinha, A., Sakib, N. and Hossain, E. (2024) 'Investigating customer churn in banking: a machine learning approach and visualization app for data science and management', <i>Data Science and Management</i> , 7(1), pp. 7–16. doi: 10.1016/j.dsm.2023.09.002.
Link	https://doi.org/10.1016/j.dsm.2023.09.002
Modelling	The study used grid-search cross-validation to investigate logistic regression, decision tree, random forest and gradient boosting models. A visualisation application was also developed to support interpretation and use of the results.
Preprocessing techniques	The study used SMOTE to address class imbalance. Customer information from different account-related variables was combined, including credit score, tenure, balance, number of products and estimated salary.
Datasets used	A Kaggle bank customer churn dataset containing 10,000 customer records. The dataset included variables such as credit score, age, tenure, balance, number of products and salary.
Key takeaways	1. Class imbalance should be considered when developing models for binary banking outcomes. This project will check the class balance during exploratory analysis and apply an imbalance-handling technique if the split is skewed, as planned in the RAAIDD log. 2. Customer account and demographic characteristics can provide useful predictive information, so the client and campaign attributes in the chosen dataset will be used as predictors. 3. Visualisation can make model findings easier to interpret and communicate, so results will be presented using clear tables and charts. Relevance to STADIOEquities: The study is relevant because identifying customers who are likely to disengage can help inform the identification of customers who may not progress from registration to activation. It also highlights the importance of addressing class imbalance in binary customer prediction problems. It concerns existing customers leaving rather than new registrants failing to activate, so it is a related but different stage of the customer journey.

Publication 3

Title	<i>Machine Learning-Based Analysis of Technology Acceptance in FinTech: A Behavioral Study Using Digital Wallet Data</i>
Harvard reference	Abbas, S.K., Hussain, M. and Rimal, Y.N. (2025) 'Machine learning-based analysis of technology acceptance in FinTech: a behavioral study using digital wallet data', <i>SN Computer Science</i> , 6, article 674. doi: 10.1007/s42979-025-04214-8.
Link	https://doi.org/10.1007/s42979-025-04214-8
Modelling	The study used logistic regression, support vector machines (SVM), random forest and XGBoost to classify high and low technology acceptance. The models were re-evaluated after applying SMOTE. SHAP and principal component analysis (PCA) were used to support model interpretation.
Preprocessing techniques	Categorical variables were label encoded, while numerical variables were standardised using z-score standardisation. Identifier columns were removed, and an 80/20 stratified train-test split was used. The dataset contained no missing values.
Datasets used	The Kaggle Digital Wallet Customer Lifetime Value (LTV) dataset, containing 7,000 records and 20 features covering demographic, transactional and behavioural information.
Key takeaways	1. Static behavioural information can make minority-class prediction difficult. 2. Temporal and session-based features may provide additional useful information. This supports the session and timing fields requested in the SS1 data request, which the chosen public dataset cannot supply. 3. SMOTE can improve the detection of minority-class observations in an imbalanced classification problem. This project will consider SMOTE or class weighting if the target is imbalanced. Relevance to STADIOEquities: The study is relevant because it applies machine learning to behavioural data within a FinTech environment. Its findings support the use of early customer behaviour as potential predictors of future activation and demonstrate the value of techniques for handling imbalanced outcomes.

Public Dataset Used for Part B

Name	Bank Marketing dataset (bank-additional-full.csv)
Harvard reference	Moro, S., Rita, P. and Cortez, P. (2014) <i>Bank Marketing</i> [Dataset]. UCI Machine Learning Repository. doi: 10.24432/C5K306.
Link	https://archive.ics.uci.edu/dataset/222/bank+marketing
Number of instances	41,188 instances in the bank-additional-full.csv file, supplied as a single file with no predefined training and test split. The older bank-full.csv version contains 45,211 instances and fewer features. The version used for this project is bank-additional-full.csv.
Number of feature columns	20 feature columns (plus the target column, y).
Number of classes	2 (yes and no).
Target range	Not applicable because this is a classification problem.
Origin	Real data from direct marketing campaigns, carried out by telephone, run by a Portuguese banking institution. The dataset is publicly available through the UCI Machine Learning Repository.

Notes on Dataset Choice and Modelling

Although the Bank Marketing dataset does not measure investment-account activation directly, its binary target represents whether a customer converts to a financial product. It therefore provides a suitable public proxy for testing the classification approach proposed for STADIOEquities.

The duration variable will be excluded during modelling because it represents information obtained during the marketing interaction and would not be available at the intended prediction point. Excluding this variable reduces the risk of target leakage and makes the modelling approach more consistent with an early-prediction scenario.